INCH POUND
MIL-P-85364A(AS)
23 September 1993
SUPERSEDING
MIL-P-85364(AS)
9 February 1981

MILITARY SPECIFICATION

PROPELLANT, CTBN 23-17

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the minimum requirements for one type of solid propellant grain (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-G-155	Graphite, Dry (For Use in Ammunition)
MIL-A-23950	Aluminum Powder, Spherical
MIL-R-82657	Resin, Epoxy, Trifunctional
MIL-A-82667/2	Ammonium Perchlorate, Conditioned

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Warfare Center Aircraft Division Lakehurst, Systems Requtemments Department, Code SR3, Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426 appearing at the end of this document or by letter.

AMSC N/A FSC 1340

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(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from DODSSP - Customer Service, Standardization Documents Order Desk, 700 Robbins Avenue, Bldg 4D, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

PUBLICATIONS

NAVAL SEA SYSTEMS COMMAND

OD 18877	Average Particle Size by the Fisher Air Permeability Apparatus
OD 18893	Water Content, Determination of
OD 18978	Particle Size Distribution by Micromerograph Analysis
OD 19554	Determination of Burning Rate Versus Pressure from PRABL

(Application for copies should be addressed to the Commanding Officer, Naval Ship Weapon Systems Engineering Station, NAVSEA Data Support Activity (Code 5H30), Port Hueneme, CA 93043-5007.)

WS 4968	Lecithin

(Application for copies should be addressed to Commanding Officer, Naval Ordnance Station, Attn: Code 802, Louisville, KY 40214.)

DRAWINGS

NAVAL AIR SYSTEMS COMMAND

1255AS128	Antioxident
1255AS137	Resin
1255AS138	Iron oxide, Monohydrate, Yellow
1255AS139	Stannous Octoate
1255AS140	Carboxyterminated Polybutadiene Acrylonitrile Copolymer

(Application for copies should be addressed to Commanding Officer, Naval Air Technical Services Facility (Code 3121), 700 Robbins Avenue, Philadelphia, PA 19111

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 203	Water Using Karl Fischer Reagent
ASTM D 792	Density and Specific Gravity (Relative Density) of Plastics by Displacement

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

CHEMICAL PROPULSION INFORMATION AGENCY (CPIA)

Publication No. 21 JANNAF Solid Propellant Mechanical Behavior Manual

(Application for copies should be addressed to Chemical Propulsion Information Agency, The Johns Hopkins University Applied Physics Laboratory, John Hopkins Road, Laurel, MD 20707.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials used in making the propellant shall conform to the specifications/drawings listed in table I.

3.2 Composition. The propellant shall contain the materials listed in table I.

3.2.1 Ammonium perchlorate (6μ). Nominal 6μ ammonium perchlorate (AP) shall be ground from the specified 200μ AP.

3.2.2 Ammonium perchlorate (45μ). Nominal 45μ AP may be ground from the specified 200μ AP.

3.3 Burning rate. The burning rate shall be determined at 0°F over the range of pressures from 1,000 to 3,000 psi. The burning rate shall be at least that calculated from the following lower limit equation:

$$r_L = 0.0230P^{0.452}$$

and not greater than that calculated from the following upper limit equation:

$$r_u = 0.0401P^{0.392}$$

At a temperature of 140°F the corresponding equation shall be:

$$r_L = 0.0276P^{0.445}$$

$$r_u = 0.0465P^{0.395}$$

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where

r_L = the lower limit of acceptable burning rate in inches per second (in/s)

r_U = the upper limit of acceptable burning rate in in/s

P = the combustion pressure in pounds per square inch absolute (psia)

NOTE: Burn rate samples shall be taken on all lots but burn rate data shall be measured only if the lot fails ballistic performance. Samples shall be held until the motor lot is placed in Condition Code A.

TABLE I. Propellant formulation.

Material	Specification/ drawing	Weight percent (nominal)	Weighout tolerance
Binder 1/ Carboxyterminated polybutadience acrylonitrile copolymer	1255AS140	14.22	±0.14
Resin, epoxy, difunctional	1255AS137	0.71	±0.01
Resin, epoxy, trifunctional	MIL-R-82657	0.99	±0.01
Stannous octoate 2/	1255AS139	0.060	±0.003
Antioxidant (Vanox 13)	1255AS128	0.620	±0.012
Lecithin	WS 4968	0.400	±0.01
Graphite	MIL-G-155	0.50	±0.01
Iron oxide, monohydrate, yellow 3/	1255AS138	0.50	±0.01
Aluminum powder, spherical, type II	MIL-A-23950	2.00	±0.04
Ammonium perchlorate 4/	5/	30.00	±0.30
6µ 45µ	MIL-A-82667/2, Type III Class 2	22.00	±0.22
200µ	MIL-A-82667/2, Type III Class 1	28.00	±0.28

1/ Binder composition may be adjusted as long as total equals 15.92.
2/ The catalyst (stannous octoate) may be adjusted from 0.04 percent to 0.08 percent, provided only the Vanox is adjusted correspondingly, for a bi-weight percent of 0.680.
3/ Iron oxide may be adjusted from 0 to 0.8 maximum for burn rate.
4/ AP adjustment blend or grind ratios may be adjusted for burn rate as long as total equals 80 percent.
5/ See 3.2.1.

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3.4 Physical properties. The physical properties of the propellant shall conform to table II (see 4.4.3).

TABLE II. Physical properties.

Property	Temperature	
	0°F	140°F
Max elongation (%) min	9	6
Max stress (psig) min	460	190
Young's modulus (psig) min	6660	3000

3.4.1 Uncured propellant. The uncured propellant shall have a maximum water (percent wt) content of 0.05 (see 4.4.4).

3.4.2 Cured propellant. The cured propellant shall have a maximum density at 25°C of 1.680 and a minimum density of 1.650 (see 4.4.5).

3.5 Workmanship. The material shall be uniform, and free from contamination, foreign material, or any other defect that would prevent its use for the purpose intended.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and test) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of section 3. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Inspection conditions. Unless otherwise specified in the contract (see 6.2), all inspections shall be performed under the following conditions:

a. Temperature: Room ambient 18°C to 25°C (65°F to 95°F)
b. Altitude: Normal ground
c. Vibration: No requirement
d. Humidity: 50 percent relative, maximum

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4.3 Quality conformance inspection. Quality conformance inspection shall consist of the test methods listed in 4.4.

4.3.1 Lot. A propellant lot shall be manufactured in one batch from single lots of ingredients. Unless otherwise specified in the contract or purchase order, a lot shall consist of not more than 4,500 pounds (see 6.2).

4.3.2 Sampling. Sampling shall be as specified in the individual test method.

4.4 Test methods.

4.4.1 6μ Ammonium percholorate.

4.4.1.1 Fisher subsieve. The particle size shall be determined in accordance with OD 18877 using a sample weight of 1.95 grams. The particle size shall be measured at the optimum porosity determined for the sample.

4.4.1.2 Micromerograph. The particle size distribution shall be determined in accordance with OD 18978 using a 45 mg sample with a 200 psi test pressure. The weight average micron value shall be calculated from the 10-point spread (95 percent - 85 percent - etc) method, wherein a summation is made of the percent weighted average particle size of each 10 percent fraction (distribution).

4.4.1.3 Surface water. The surface water (percent wt) shall be determined in accordance with OD 18893 using a 10 gram sample dissolved in 100 mL of methanol pre-saturated with 15 grams of ground AP.

4.4.2 Burning rate. The burning rate shall be determined in accordance with OD 19554 using a 1- by 2.60- by 6.00-inch slab.

4.4.3 Physical properties. The test for physical properties shall be in accordance with section 4 of CPIA Publication NO. 21.

4.4.4 Water. The water (percent wt) of the uncured propellant shall be determined in accordance with ASTM E 203 using a 10- to 20-gram sample in 100 mL of 3:1 chloroform/methanol solution. Allow the sample to stir for 10 minutes and run a timed blank on the neutralized carrier.

4.4.5 Density. The density of the cured propellant shall be tested in accordance with ASTM D 792 using a 1-inch cube sample and hexane as the immersion liquid.

5. PACKAGING

This section is not applicable to this specification.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The propellant described herein is intended for use in the Mk 23 Mods 2 and 3 Rocket Motor.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

c. Inspection conditions, if other than as specified (see 4.3).

d. Lot size, if other than as specified (see 4.3.1).

e. That the safety precaution requirements of the "Contractors' Safety Manual for Ammunition and Explosives," DOD 4145.26M, are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore," OP 5, are applicable.

6.3 Subject term (key word) listing.

MK 23 Mod 2 Jato
MK 23 Mod 3 Jato

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - AS

(Project 1340-N848)

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INSTRUCTIONS

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-P-85364A(AS)	93/9/23

3. DOCUMENT TITLE
PROPELLANT, CTBN 23-17

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
COMMANDING OFFICER, NAVAL AIR WARFARE CENTER AIRCRAFT DIVISION LAKEHURST SYSTEMS REQUIREMENTS DEPARTMENT	(1) Commercial (908) 323-7488	(2) AUTOVON 624-7488
c. ADDRESS (Include Zip Code) CODE SR3 LAKEHURST, NJ 08733-5100	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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